

`-d` helps Jenkins to run in a detached mode, where even if we terminate the session, it will not exit.

`-p 8080:8080` maps the 8080 port of the Docker container to the 8080 port on the host machine. The first number is the host machine's port number, and it can be set to any other port number.

`-v` options assign volumes to the Jenkins container. Volumes provide persistent storage that exists independently from containers but can be attached to one or more containers, as necessary.

Finally, `jenkins/jenkins` is the name of the image we want to use.

- Similarly, there is the Jenkins Blue Ocean image available to build the pipeline with pipeline as code using Blue Ocean.

```
docker run -d -p 8080:8080 -v $PWD/
jenkins:/var/jenkins_home -t jenkinsci/
BlueOcean
```

- To get a bash/ssh inside a running container, we can use this command:

```
docker exec -i -t jenkins/jenkins:its
bash
```

- To check the status of containers, use the following command:

```
docker ps -a
```

The above procedure was about running Jenkins on Docker. We can also do Jenkins integration with Docker (installed on machine).

- Go to the Jenkins dashboard. Click on *Manage Jenkins > Manage plugin > Install Docker plugin*.
- Then go to *Configure system > Add a new cloud > Docker*.
- Add a Docker template to an agent node. Here, we can add a list of images that can act as slaves.

→ **Labels:** This is the name of the agent node.

→ **Docker image:** This provides the image name.

→ **Remote filing system root:** This is Jenkins' home path in the container.

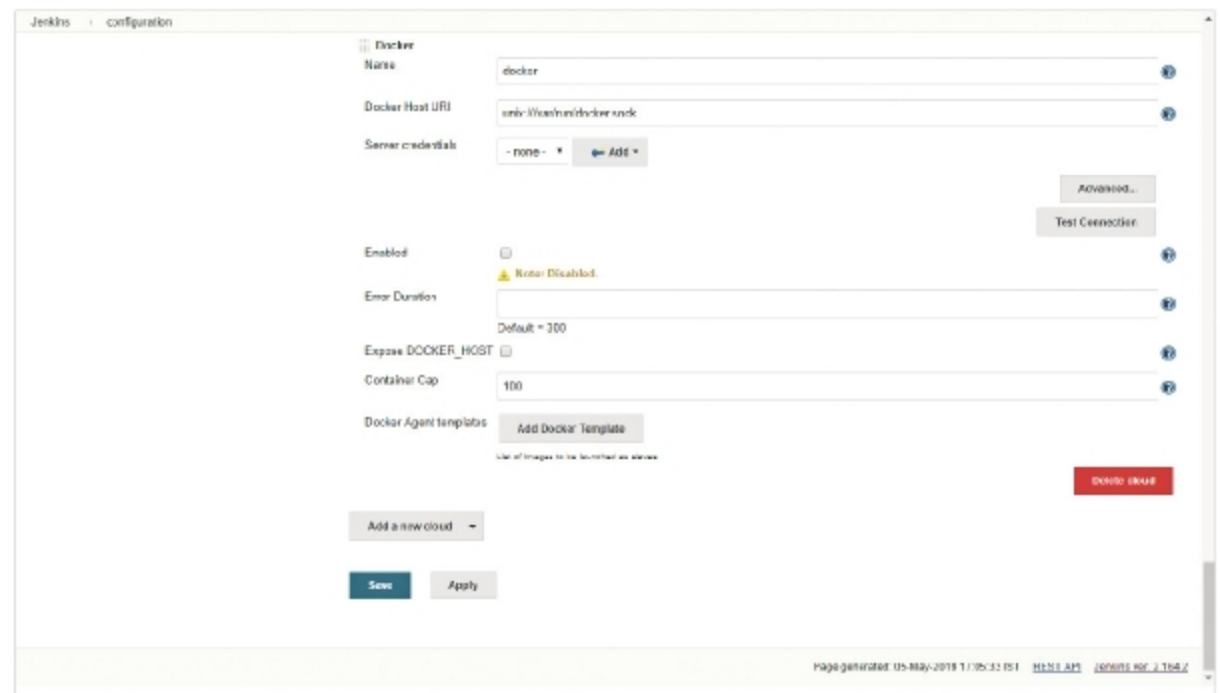


Figure 9: Docker cloud

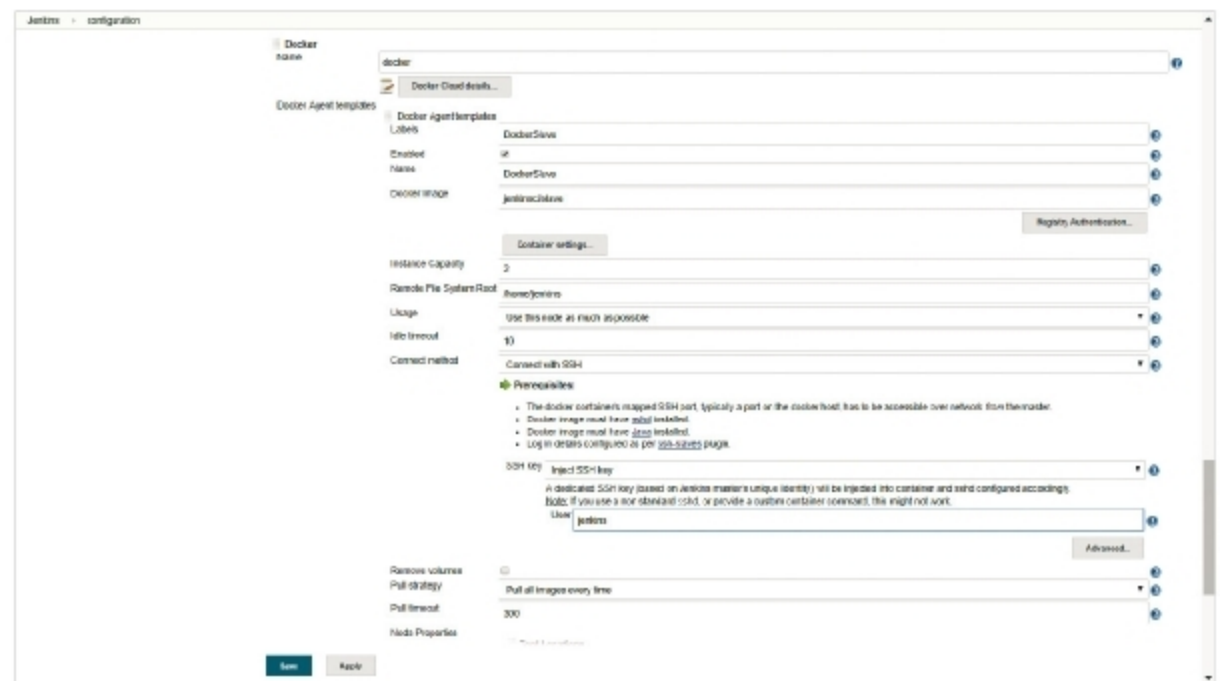


Figure 10: Docker template

→ **Connect method:** Selects the access method to build our image as 'Connect with SSH'

- Save the template.
- The job can be configured as

'restrict to run' on the label of the agent template.

We have now created a Jenkins slave in a Docker container, which is portable and lightweight. 

References

- [1] Jenkins 2.x Continuous Integration Cookbook - Third Edition (<https://www.amazon.com/gp/product/B0751HZ96H>)
- [2] Jenkins Essentials - Second Edition (<https://www.amazon.com/gp/product/B073PGCKHY>)
- [3] DevOps Bootcamp: The fastest way to learn DevOps (<https://www.amazon.com/gp/product/B07259Y9M8>)
- [4] DevOps for Web Development (<https://www.amazon.com/gp/product/1786465701>)

By: Sneha Kavade and Mitesh Soni

Sneha Kavade is a DevOps enthusiast, and is eager to learn new technologies related to application life cycle management automation. Mitesh Soni is the author of the book 'Implementing DevOps with Microsoft Azure'. He contributes occasionally to <https://clean-clouds.com> and <https://etutorialsworld.com>.